

UX/UI

For Researchers

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Intro

My work

[Word Lapse](#)

[Preprint Similarity Search](#)

[Het.io](#) and [Connectivity Search](#)

[MyGeneset.info](#)

[Adage](#)

[Manubot](#)

[Monarch Initiative](#)

[Simplex](#)

[Human Microbiome Compendium](#)

[Lab Website Template](#)

... and many other [personal projects](#)

Overview



Content

Tenets and advice

Common researcher pitfalls

Standard paradigms

Examples and exercises

~~HTML/CSS/JS~~



Takeaways

Better idea of how to think like a designer

Handful of tips to put into practice

Appreciation and respect for the craft

UX vs. UI

User Experience

 The *feel*

Information organization

User workflows/paths

General behavior

Research-focused

High level (site, pages)

User Interface

 The *look*

Visual consistency and meaning

Polish, "pixel-perfect"

General aesthetics

Design-focused

Low level (components)

UX Process

Do research



User research: **interviews**, surveys, a/b tests, card sorting, focus groups, and many more

What do you like/dislike? Pain points?
Common tasks? Feature wishlist?
Watch them use apps.



Market research: direct competitors, sites in similar domain, popular sites

Use other apps as a user. What do they do well? What is painful? What ideas can you take?



Document and **judiciously** incorporate

Users are good at identifying problems, but not solving them

Create user stories



Plots a specific user **journey** through the app



Answers the questions: **who**, **what**, **why**, and **how**



Helps define important **use cases** and **goals**

I am a biologist researching genes of a particular species. I'm not a programmer. I go to mygeneset.info. I want to search for genes with particular names, but filter by a specific taxon. I want to save them to a named collection. Finally, I want to export them in a format where I can map them to my own annotations by ID.

Organize your ideas

Clean organization → intuitive site



Come up with a simple **mental model**

Could a layperson get the basic idea?
E.g. inputs vs. outputs.



Categorize your content in large to small chunks

What is truly important? What are the first things the user should see?

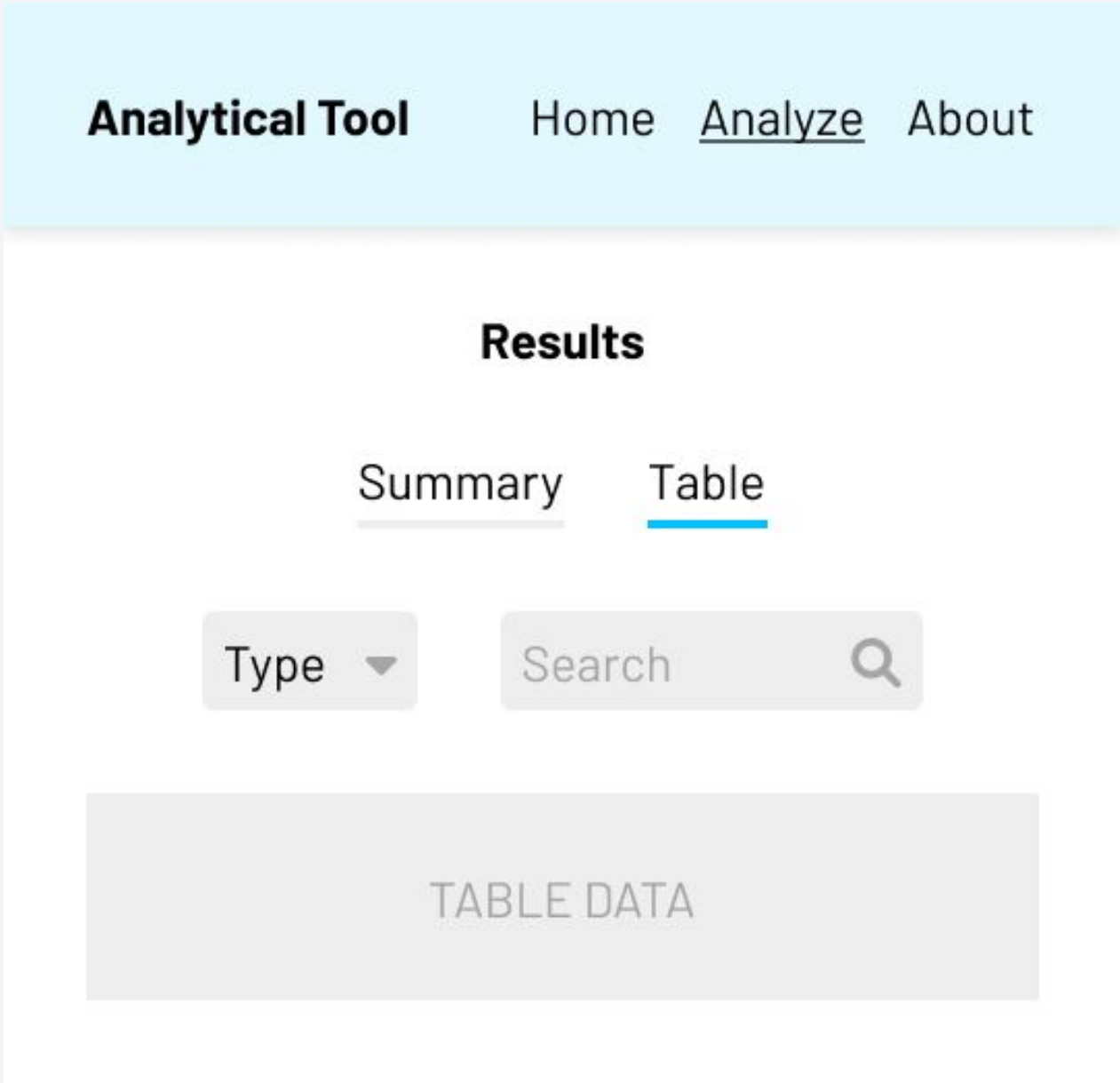


Stick to the categories and frequently **re-evaluate**

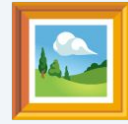
Don't let things become a ball of duct tape

Organize your ideas (example)

- Ecosystem:** Some Analytical Suite
- Site:** Some Analytical Tool
- Pages:** Home Analyze About
- Sections:** Inputs Results
- Tab Group:** Summary Table
- Control Group:** Filters Data Actions
- Controls:** Type Search



Make a sketch



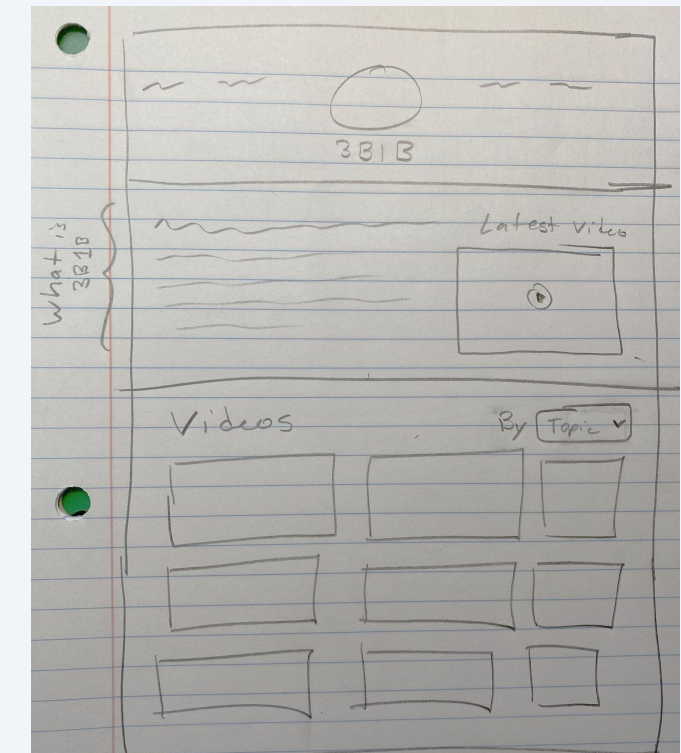
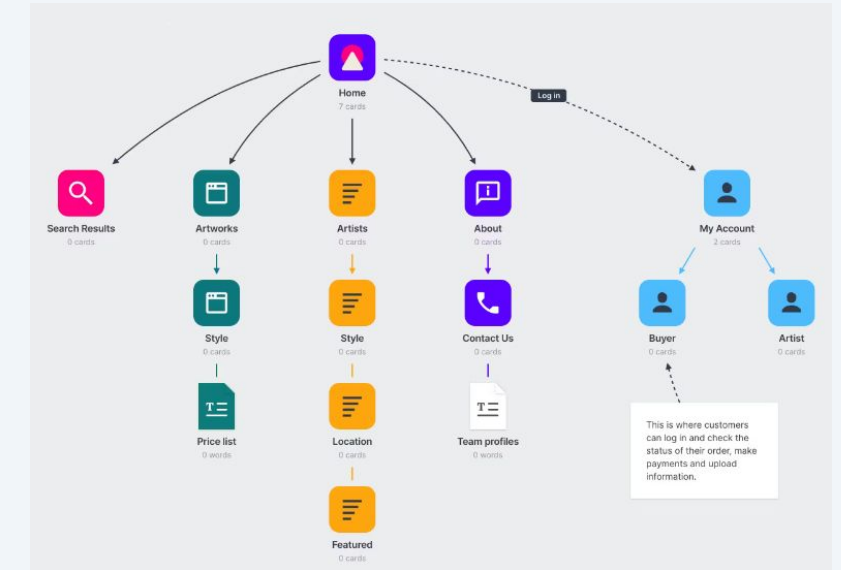
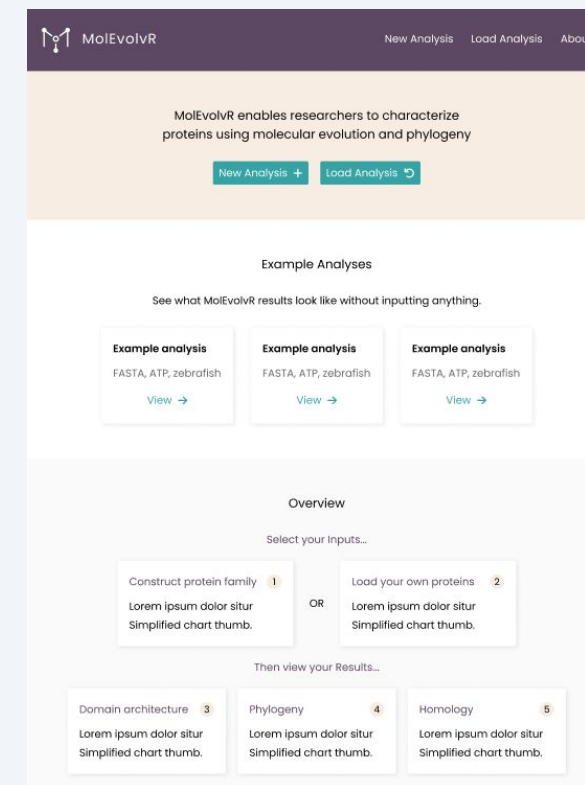
Design should **inform** implementation, not the inverse



Draw and **map** your ideas first. Coding comes later.



Try Figma, Lucidchart, or pencil & paper



UX Principles

Keep it simple



Don't overthink it.
Don't overanalyze.

You're deep in it, the user isn't.
Beware of tunnel vision.



Use **progressive disclosure**.
Limit options and choices.

Display what's truly essential, hide what isn't. Beware analysis paralysis.



Stick to **established patterns**.
Don't get *too* creative.

Have you ever seen a site do something like this before?

Show, don't tell

"A UI is like a joke: if you have to explain it, it's not that good."



Favor simple **visuals**. Avoid verbose **text**.



Use spacing, flow, and **context** to **disambiguate**



Use **examples**!

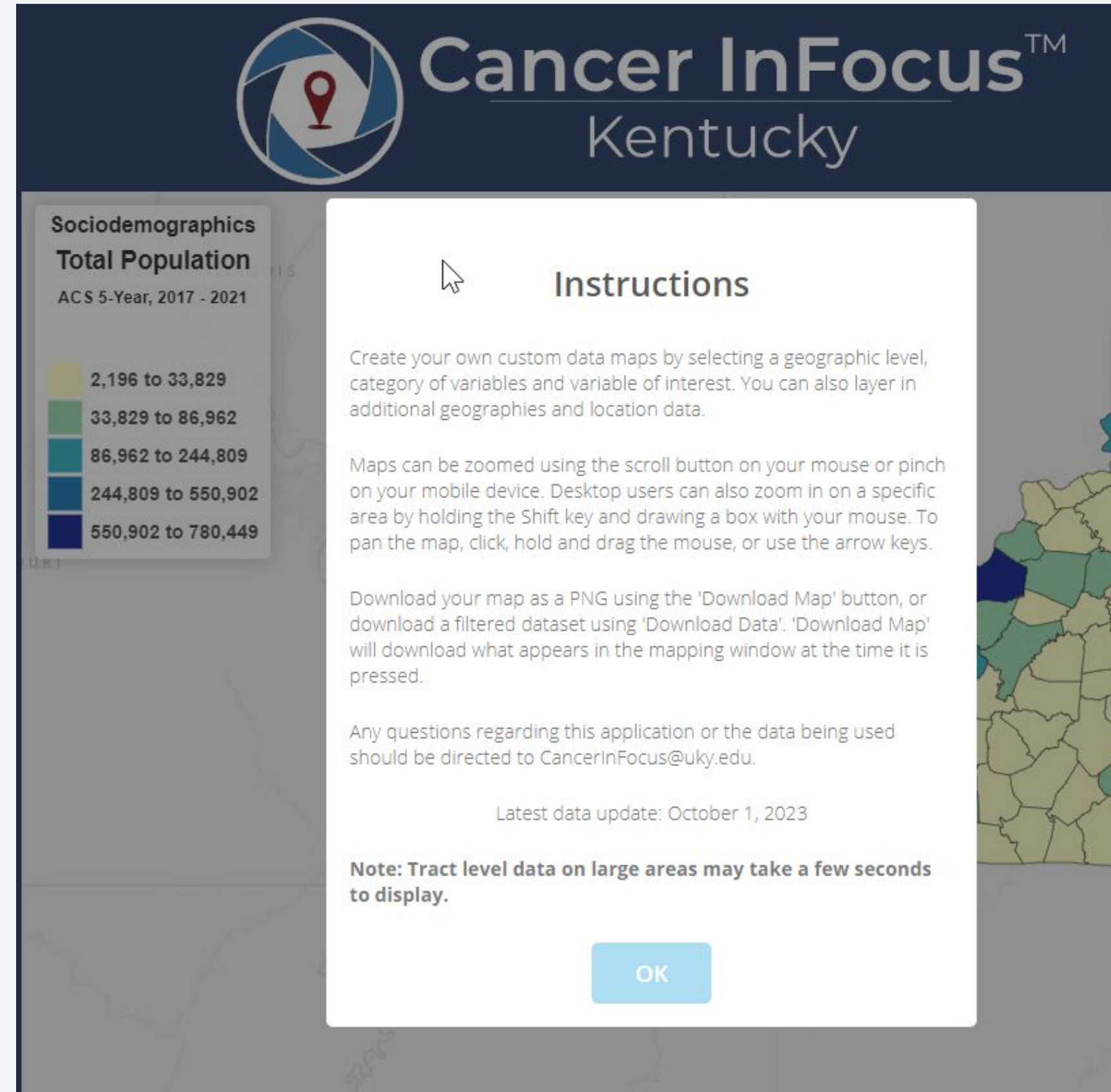
First, acquire your eggs. Then you can crack the eggs into a hot pan to fry them.



How to cook eggs:

- 1 Buy
- 2 Crack
- 3 Fry
- 4 Enjoy!

Show, don't tell (cont.)



Focus on copywriting

"Why waste time say lot word when few word do trick?" – Ancient proverb



Ruthlessly **prune** text

~~Use the minimum amount of words to convey the most essential ideas~~



Use **consistent** structure and terminology

Parts of speech. Passive vs. active. Granularity.



Use **simple** words and phrases

Avoid long sentences, college-level words, idioms. Use natural language.

UI Principles

Layout is key



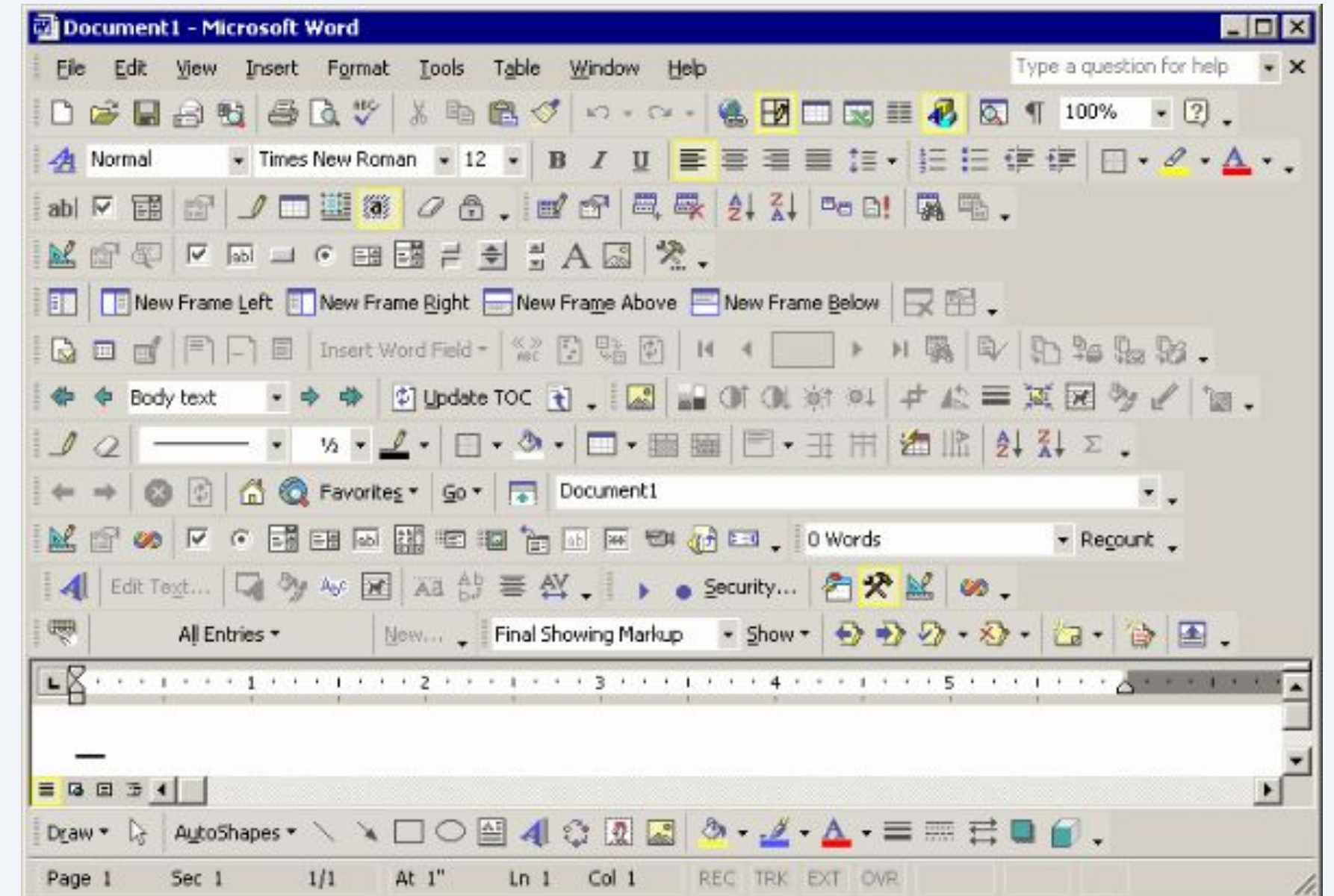
Let it breathe! Blank space is your **friend**.



Keep the **flow** simple and clear. Embrace linearity. **One thing** at a time.



Not all **screens** are desktops! Responsive design.



Establish a visual language



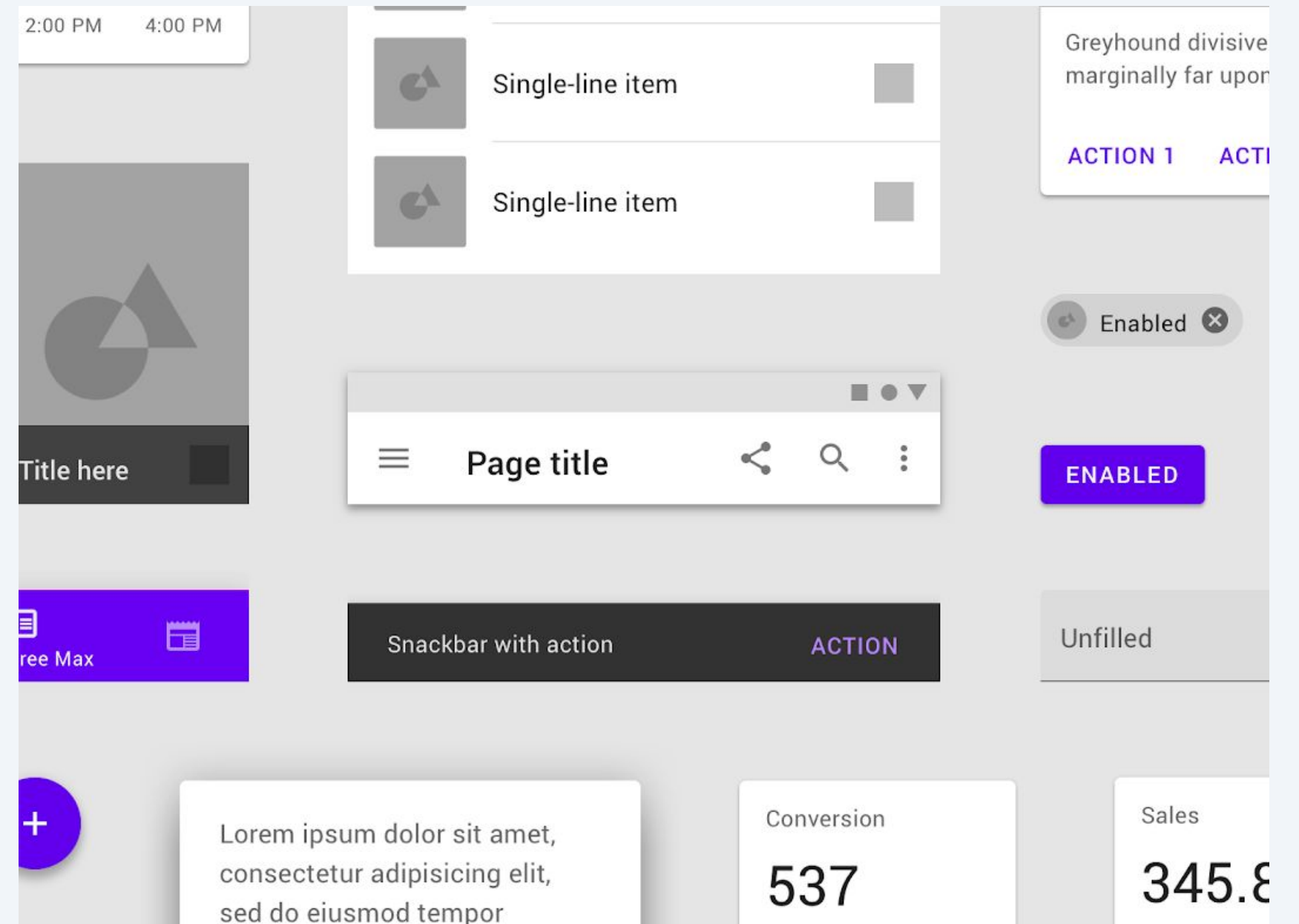
Icons, colors, shapes, fonts, weights, sizes, order, position, spacing, language, etc.



Make a **palette** to enforce consistency



Keep in mind **common associations** and expectations



Accessibility

What is accessibility?

Partial vision disabilities

Color blind (7+ different kinds!)

Motor disabilities

Cognitive disabilities

Hard of hearing

Communication disorders

Psychological disabilities

Motion sickness

Seizure disorder

Physical pain

Respect accessibility. Not everyone is like you.

Common accessibility issues



No keyboard focus



No alt-text, CC



Non-semantic HTML



No headings



Bad screen reader support



Low contrast, small text



Color-reliant



Difficult click targets



Convoluted language



Flashing, motion

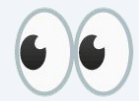
Use **Lighthouse** (in Chrome dev tools) or **Deque Axe** directly

Tips

Pretty tips



Clean → pretty



Bold to emph., light de-emph.



Hero section



Icons next to text



Line-spacing



Logo



Reference (steal)



[Tailwind](#)

[Material UI](#)

[Coolors](#)



[Google Fonts](#)

[GH Fonts](#)

[FontSpace](#)



[Font Awesome](#)

[Material Icons](#)

[React Icons](#) (meta-collection)

Tech tips



Save data to local device



Persist state in URL



Auto-populate from other pages



Auto-scroll to sections



Show extra info in tooltips



CSS position sticky



Localization utility functions



<https://some-tool.org/analysis/abc123?tab=results>

Power to raise step ranges by. Only affects which colors are assigned to which values.

= 1 is linear

> 1 exaggerates differences at low values

< 1 exaggerates differences at high values

Scale power

2.25

```
(12345.67).toLocaleString()
```

12 345,67



Exercise

Ridley Creek Scavenger Hunt

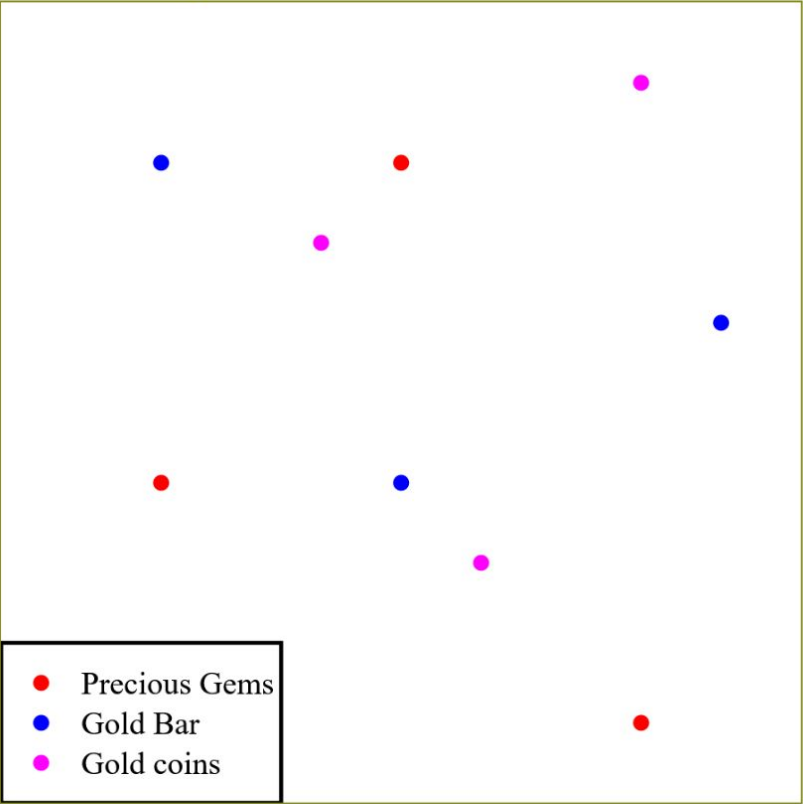
[2010-2020 Hunts](#) [2021 Hunt](#) [2022 Hunt](#) [2023 Hunt](#)

Get ready for the 12th annual Ridley Creek Treasure hunt on June 31st, 2023.
Begins at 12:00 noon.

Need help? See FAQs below

Click here to reveal a hint on the map: [?](#)

Treasure map:



Show only: ☐ Gems ☐ Gold Bars ☐ Coins
☐ Show all treasures

[Download map to device](#)

[Download w/ selected gems](#)

[Join our newsletter for updates!](#)

FAQs

How to show only certain types of treasure

You can use the dropdown menus and checkbox above to either choose a single type of treasure to show on the map, or optionally show all of the treasures at the same time.

How do I look for jewels?

You can find treasures by using the above map to locate items in the real world. The map corresponds to a region between -75.47184182166482 and -75.42030782388845 longitude and 39.93645851286192 and 39.965309814844034 latitude, around Ridley Creek State Park.

Tips

It's more fun if you don't use the hints or ask people for help. You should try your best to find the treasures on your own. Don't give up, you can go the whole nine yards!

Exercise (cleaned up)

Real map, real treasure! Get ready for the fun!

📍 WHERE [Ridley Creek State Park](#)
📅 DAY June 31st, 2023
🕒 WHEN 12:00 noon — 3:00 pm

Map

Show: [All Treasures](#) ▾



Download

Hint

Tips

1. Do it yourself, try not to use hints!
2. Study the map, you can do it!
3. Have fun!

Stay up to date

Join our newsletter

Questions